

DATA SCIENCE RESOURCES

Agriculture & Environment Projects

Project 1. Crop Yield Prediction Using Weather and Soil Data

Problem Statement: Farmers face uncertainty in predicting crop yields due to variable weather, soil quality, and irrigation conditions. Inaccurate yield forecasts can impact food supply chains, market pricing, and agricultural planning.

Objective: Build a supervised learning model that predicts crop yield (in tons/hectare) using features like soil type, rainfall, temperature, and fertilizer usage.

Dataset Source: [India's District-Level Crop Yield Dataset](#)

Techniques & Concepts

- Regression modeling (Linear, Random Forest, XGBoost)
- Feature scaling and encoding (categorical soil types)
- Handling multicollinearity
- Cross-validation for model reliability

Tools & Libraries: Python, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn

Expected Outcome: An ML model that provides yield predictions for specific crops based on region, soil, and climate inputs, useful for farmers, agri-tech startups, and policy makers.

Project 2. Weather Condition Classifier from Sensor or Image Data

Problem Statement: In remote farming areas or automated systems, it is often necessary to classify weather conditions (sunny, rainy, foggy, etc.) to make decisions about irrigation, pesticide use, or crop storage.

Objective: Create a classification model that predicts weather condition categories using structured sensor data or images (e.g., satellite or sky cam images).

Dataset Source: [Weather Classification Dataset](#)

Techniques & Concepts

- Classification (Logistic Regression, SVM, CNN for images)
- Data augmentation for image input
- Multi-class confusion matrix
- Model evaluation with accuracy, precision, recall

Tools & Libraries: Python, Pandas, Scikit-learn, TensorFlow/Keras (for image-based), Matplotlib

Expected Outcome: A classifier that predicts weather types based on given input — ideal for use in automated agriculture platforms or smart irrigation systems.

Project 3. Forest Fire Detection Using Remote Sensing Data

Problem Statement: Forest fires cause massive ecological damage and are difficult to detect early. Automating fire detection from satellite imagery or sensor data can improve response times and reduce loss.

Objective: Build a model that detects potential forest fires from structured environmental data (e.g., temperature, humidity, wind) or satellite images.

Dataset Source: [Forest Fire Dataset \(UCI ML Repository\)](#)

Techniques & Concepts

- Binary classification (fire vs no fire)
- Logistic Regression, Random Forest, or CNNs for image classification
- ROC-AUC curve analysis
- Geospatial heatmap visualization

Tools & Libraries: Python, Scikit-learn, GeoPandas, TensorFlow/Keras (for image input), Matplotlib, Folium

Expected Outcome: A working model or dashboard that predicts fire risk or detects fires based on real-time or historical data, ideal for use in forestry and disaster response agencies.
